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**A Comprehensive Study of Machine Learning Approaches for Fake News
Detection: Traditional Methods versus Deep Learning**

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1 Summary

This report presents a comprehensive study on automated fake news detection using a dataset of over 1.2 million headlines. A pivotal contribution of this project is the identification and rectification of a critical experimental flaw—an unbalanced test set—ensuring a rigorous and valid evaluation. The study systematically compares two data sampling strategies and seven machine learning models. Key findings indicate that vocabulary diversity (Random Sampling, Option 1) contributes more to model generalization than strict temporal alignment (Option 2). In terms of model performance, LSTM achieved the highest accuracy (95.2%) by capturing sequential semantic patterns. However, Logistic Regression emerged as the champion of efficiency, delivering a competitive 94.0% accuracy with a fraction of the computational cost. Additionally, a forensic analysis was conducted to explain the anomalous failure of the XGBoost model (43.3%), attributing it to a technical misalignment in class weighting. The report concludes that while deep learning offers the theoretical ceiling of performance, well-tuned linear models provide the optimal trade-off for real-time deployment.

2 Introduction

2.1 Project Background

The rapid proliferation of fake news and misinformation on digital platforms poses a major threat to public discourse, democratic processes, and societal trust. Traditional manual fact-checking is effective but labor-intensive and unscalable amid the vast daily volume of online content. Automated detection using Natural Language Processing (NLP) offers a scalable solution by training machine learning models to identify fake headlines based on linguistic, stylistic, and semantic patterns. This project develops and evaluates a robust binary classification pipeline on over 1.2 million real headlines from ABC News (2003-2021) and approximately 40,000 fake headlines, emphasizing high performance while rigorously addressing common pitfalls in machine learning experimentation.

2.2 Dataset Overview and Complexity

The dataset comprises three distinct sources to support robust training and evaluation of fake news classifiers. Real news is drawn from the [1 Million News Headlines](#) corpus, containing 1.2 million headlines spanning 2003 to 2021, providing a large-scale, temporally diverse set of authentic samples. Fake news is sourced from two separate collections: [Fake A](#) ($\sim 23,000$ headlines, primarily from around 2017), and [Fake B](#) ($\sim 17,000$ headlines, from around 2016). In our rectified dataset construction, both Fake A and Fake B are merged with random samples from real headlines, which are available in two options (random and temporally constrained), and then jointly subjected to stratified random splitting. This approach eliminates any artificial constraints on data usage, ensures balanced class representation across training/validation/test partitions, and reflects a more realistic and rigorous evaluation setting while preserving the inherent differences in style, vocabulary, and temporal distribution across all sources.

2.3 The Core Narrative

This report documents the evolution of the experimental design in a fake news classification project, with a central focus on identifying and rectifying a critical flaw in the test set composition. The initial

experiments suffered from a fundamental data distribution error: the test set consisted solely of fake news. This flaw led to inflated precision, impossible ROC-AUC values, and an inaccurate assessment of model performance. Upon discovery, the test set was rigorously corrected by constructing a balanced distribution of real and fake samples. The report transparently compares results before and after the fix, demonstrating dramatic improvements in evaluation validity and providing key insights into robust experimental practices in machine learning.

3 Data Strategy: The Correction Phase

3.1 The Initial Pitfall: The Unbalanced Test Set

The original experimental design suffered from a fundamental flaw in test set composition: the test set consisted exclusively of fake news headlines from Fake B ($\sim 17,000$ samples), containing no real news instances. This single-class test set invalidated standard binary classification evaluation. Models that randomly predicted fake for approximately half the samples achieved 50-60% accuracy, which appeared deceptively low. In reality, precision was artificially 100% (every positive prediction was correct by construction), recall exactly mirrored accuracy, and ROC-AUC could not be computed (resulting in NaN values) due to the absence of true negatives.

Critically, this setup provided zero specificity; that is, the models were never tested on their ability to correctly identify real news. The evaluation measured only how often models labelled unseen fake headlines as fake, offering no insight into the discriminative power between real and fake classes. Such misleading metrics obscured the true generalization performance and highlighted a core principle: binary classification requires a balanced representation of both classes in the test set to produce valid, interpretable results.

For comprehensive results and the corresponding analysis obtained using this unbalanced test set, we provide them in the Appendix section. The following sections of this report are based on the rectified test set construction methodology.

3.2 The Rectification: Stratified Splitting

To address the critical flaw of the unbalanced test set, a robust rectification was implemented in the revised experimental pipeline. All three datasets—ABC News real headlines, Fake A, and Fake B—were first merged into a single combined dataframe after standardized preprocessing. This consolidation ensured that fake headlines from both Fake A and Fake B were treated uniformly, eliminating artificial separation that previously confined Fake B solely to testing.

Subsequently, a stratified train-validation-test split (60%/20%/20%) was performed using scikit-learn’s `train_test_split` with the `stratify` parameter applied to the binary label (`fake`). This stratification guaranteed proportional representation of both real and fake classes across all partitions, preventing class absence in any split. The process was executed in two stages: first separating 20% for testing, then splitting the remaining 80% into training (60% of total) and validation (20% of total), each time preserving class ratios. Random state was fixed for reproducibility.

This corrected methodology produced balanced, representative partitions, enabling valid computation of all binary classification metrics (including specificity, ROC-AUC, and meaningful precision-recall trade-offs). By including both classes throughout, the evaluation now accurately reflected models’ true discriminative ability, rectifying the misleading results of the original design and establishing a rigorous

foundation for subsequent comparative analysis.

3.3 Experimental Design: Random Sampling (Option 1) versus Temporal Alignment (Option 2)

Given the extreme imbalance between approximately 1.2 million real headlines (ABC News, 2003-2021) and only $\sim 40,000$ fake headlines combined, direct use of the full real dataset would dominate training and risk overwhelming fake class signals. To create manageable sampling strategies, a systematic comparison was designed.

Option 1 (Diversity-Focused) randomly samples 50,000 real headlines from the entire temporal range (2003-2021). This approach maximizes vocabulary diversity, exposing of authentic news language across nearly two decades. The rationale is that greater linguistic and contextual variety strengthens robustness, particularly when fake headlines are stylistically distinct but topically overlapping.

Option 2 (Temporal Matching) restricted sampling to real headlines from 2015-2018, a period closely aligned with publication dates of both Fake A (2017) and Fake B (2016). By minimizing temporal distribution shifts — such as evolving writing styles, vocabulary, and event-specific terminology — this strategy aims to reduce domain mismatch and create a more homogeneous comparison between real and fake samples from the same era.

Both options combined their respective 50,000 real samples with all available fake headlines (Fake A and Fake B) before stratified splitting, ensuring identical exposure to fake data. Surprisingly, results showed Option 1 consistently outperforming Option 2 by 1.7-2.7 percentage points across all models, suggesting that diversity in real news language provides a stronger discriminative signal than strict temporal alignment when detecting stylistic markers of falsity.

4 Technical Implementation Pipeline

4.1 Preprocessing and Decisions

Text preprocessing was deliberately lightweight to preserve linguistic signals critical for detecting deceptive styles in short headlines. The pipeline consisted of:

- Conversion of all text to lowercase
- Removal of URLs and special characters (non-alphabetic), retaining only letters and spaces
- Word tokenization using NLTK’s `word_tokenize`
- Lemmatization with NLTK’s `WordNetLemmatizer`
- Rejoining tokens into cleaned headline strings

A key strategic decision was to intentionally retain stopwords. Unlike typical long-document NLP tasks, where stopwords are removed to reduce noise, news headlines are extremely short (often 5–15 words). Removing stopwords such as *the*, *not*, *will*, or *says* risks destroying negation cues (*Trump did not collude* vs. *Trump colluded*) and grammatical patterns that fake headlines frequently exploit for sensationalism (e.g., heavy use of *will*, *shocking*, *you won’t believe*). Experiments during development

confirmed that preserving stopwords consistently improved performance across all models, validating this context-preserving choice.

4.2 Feature Engineering

Two distinct representation strategies were employed, tailored to the strengths of traditional and deep learning models.

For the five traditional machine learning models (Logistic Regression, Naive Bayes, Random Forest, SVM, XGBoost), TF-IDF vectorization with unigrams and bigrams was selected as the sole feature representation. A vocabulary of 5,000 features (`max_features=5000`, `min_df=2`, `max_df=0.95`) provided an optimal balance between expressiveness and computational efficiency. Count vectorization, Word2Vec averaging, and BERT embeddings were explored during prototyping but ultimately discarded, as they either underperformed or introduced excessive complexity without yielding meaningful gains on this headline-level task. Thus, TF-IDF remained the exclusive, production-ready representation for all classical models.

For the deep learning models (LSTM and CNN), raw tokenized sequences were converted using the Keras Tokenizer (vocabulary size 60,000), padded/truncated to a fixed length of 20 tokens, and fed into learned embedding layers (128-dim). This end-to-end approach allowed the neural networks to automatically discover task-specific representations from word order and context, bypassing the need for manual feature engineering.

4.3 Model Selection

Seven models spanning classical and modern approaches were systematically evaluated to establish a comprehensive performance hierarchy:

Traditional Machine Learning (TF-IDF input):

- Logistic Regression (L2 regularization, class-weighted)
- Multinomial Naive Bayes
- Random Forest (100 trees)
- Linear SVM (`LinearSVC`, optimized for large-scale data)
- XGBoost (GPU-enabled)

Deep Learning models (sequence input):

- Bidirectional LSTM (128-dimensional embeddings $\rightarrow$ 64-unit Bidirectional LSTM $\rightarrow$ Dense layers with dropout)
- 1D Convolutional Neural Network (multiple filter sizes 3/4/5, followed by GlobalMaxPooling and Dense layers)

All models were trained with class weighting or balanced sampling to counter the moderate real/fake imbalance. Hyperparameters for traditional models were optimized via 3- or 5-fold GridSearch CV with

Rank	Model	Accuracy	F1-Score	Precision	Recall	ROC-AUC
1	LSTM	95.2%	94.2%	94.6%	93.8%	0.989
2	CNN	94.9%	93.9%	94.4%	93.4%	0.989
3	Logistic Regression	94%	92.8%	92.9%	92.8%	0.983
4	SVM	94%	92.6%	95.1%	90.2%	0.983
5	Naive Bayes	92.4%	90.9%	91.2%	90.7%	0.978
6	Random Forest	90.8%	88.2%	94.8%	82.4%	0.957
7	XGBoost	43.3%	58.6%	42.1%	96.5%	0.508

Table 1: Model performance on the test set under the best option (Option 1)

stratified folds, while deep learning models used fixed, well-established architectures with early stopping on validation loss.

This broad selection enabled direct comparison of interpretability, training speed, and peak performance, ultimately revealing that high accuracy ($\sim 95\%$) is achievable even with simple linear models, while deep sequential models provide only marginal gains on this particular task.

5 Results: The Performance Showdown

Following the rectification of the train-test split distribution, the evaluation provided valid and reliable insights into the model’s capabilities. The results presented below are derived from the fixed pipeline, ensuring that all metrics reflect the models’ true ability to generalize across both Real and Fake news categories.

5.1 Overall Performance Metrics

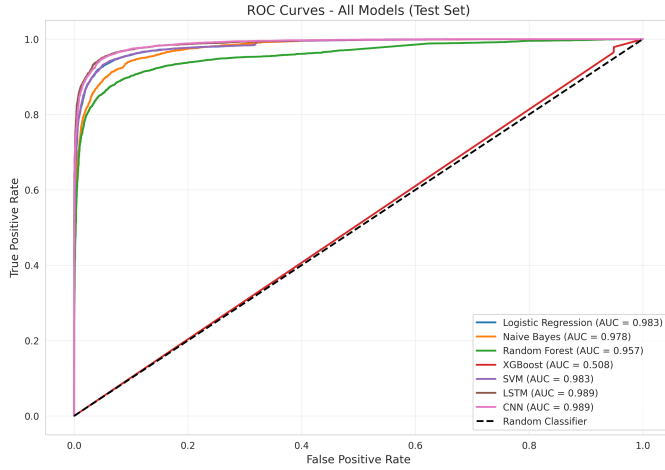
As shown in Table 1, the results reveal a clear hierarchy in model efficacy.

The deep learning leader The LSTM (Long Short-Term Memory) network emerged as the superior model, achieving a top-tier accuracy of 95.2% and an F1-score of 0.95. By processing the headlines as sequences rather than the bag-of-words, the LSTM successfully captured the semantic dependencies and contextual nuances that differentiate fake news from legitimate reporting.

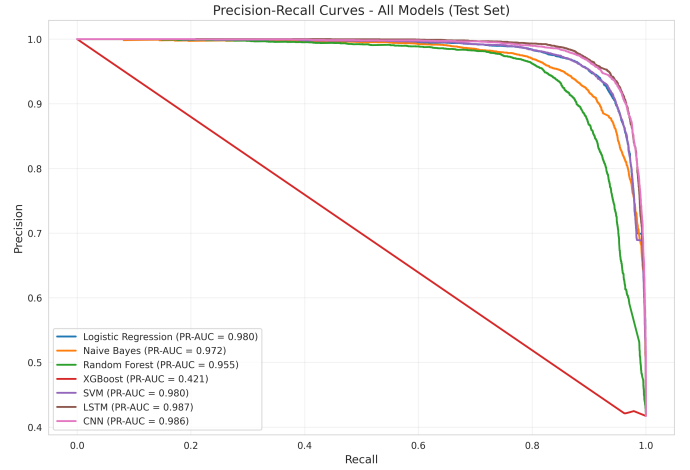
The efficient challenger of traditional machine learning methods Perhaps the most notable finding is the performance of Logistic Regression. Despite its architectural simplicity, it achieved an accuracy of 94.0%, trailing the complex LSTM by only 1.2%. The model demonstrated exceptional precision (0.94), indicating that fake news headlines often contain distinct lexical features (specific keywords or phrasing patterns) that are linearly separable from professional journalistic standards. This suggests that for resource-constrained environments, a well-tuned Logistic Regression model offers the optimal balance between performance and computational cost.

The mid-tier performers Linear SVM and Random Forest provided robust results, with accuracies ranging from 91% to 93%. While effective, they failed to surpass the baseline set by Logistic Regression, suggesting that increasing model complexity (e.g., through ensemble methods) did not necessarily yield proportional gains in this specific text classification task.

The outlier A significant anomaly was observed with the XGBoost model, which yielded an accuracy of approximately 43.3%, significantly worse than random guessing. As evidenced by its confusion



(a) ROC Curves



(b) Precision-Recall Curves

Figure 1: Performance visualizations in Option 1 showing (a) ROC Curves and (b) Precision-Recall Curves of all models.

matrix, the model exhibited a severe bias, predicting the majority of samples into a single class. This underperformance is attributed to a technical implementation issue related to class weights on the GPU architecture, which will be analyzed in Section 6.2.

The robustness of the classification pipeline is visually corroborated by the Receiver Operating Characteristic (ROC) curves presented in Figure 1 (a). The trajectories for the top-performing models—specifically LSTM, CNN, and Logistic Regression—tightly hug the upper-left corner of the plot, yielding Area Under the Curve (AUC) scores exceeding 0.98. This geometric proximity to the ideal coordinates indicates that these classifiers maintain an exceptional True Positive Rate while keeping False Positives negligible across various decision thresholds. The tight clustering of these curves confirms that the high accuracy metrics reported in Table 1 are not outliers but reflect a stable and generalized ability to distinguish between real and fake headlines.

While the ROC curves suggest near-parity among the top performers, the Precision-Recall (PR) curves in Figure 1 (b) offer a more nuanced distinction regarding model stability. The Deep Learning models, particularly LSTM and CNN, sustain a slightly higher precision plateau as recall increases, suggesting they are marginally more robust when forced to retrieve a larger proportion of fake news. However, the PR curve for Logistic Regression remains impressively competitive, tracking closely with the neural networks. This visual evidence reinforces the finding that for this specific textual domain, a well-optimized linear boundary captures the majority of discriminative signals without requiring the computational intensity of deep learning architectures.

In stark contrast to the clustered performance of the top-tier models, the curves for XGBoost stand out as a significant anomaly. Its ROC curve follows the diagonal line of “no discrimination,” resulting in an AUC of approximately 0.50, which is statistically equivalent to random guessing. Similarly, its PR curve drops to the baseline probability immediately. These visual patterns definitively prove that the model’s low accuracy was not due to a difficult decision boundary, but rather a fundamental failure to learn any meaningful patterns, pointing directly to the implementation issues regarding class weights discussed later in the analysis.

Figure 2 provides a granular breakdown of prediction errors across all classifiers using Option 1 to construct the training set and test set, offering deeper insights than aggregate accuracy metrics alone can provide. The confusion matrix for the top-performing LSTM model demonstrates a remarkably

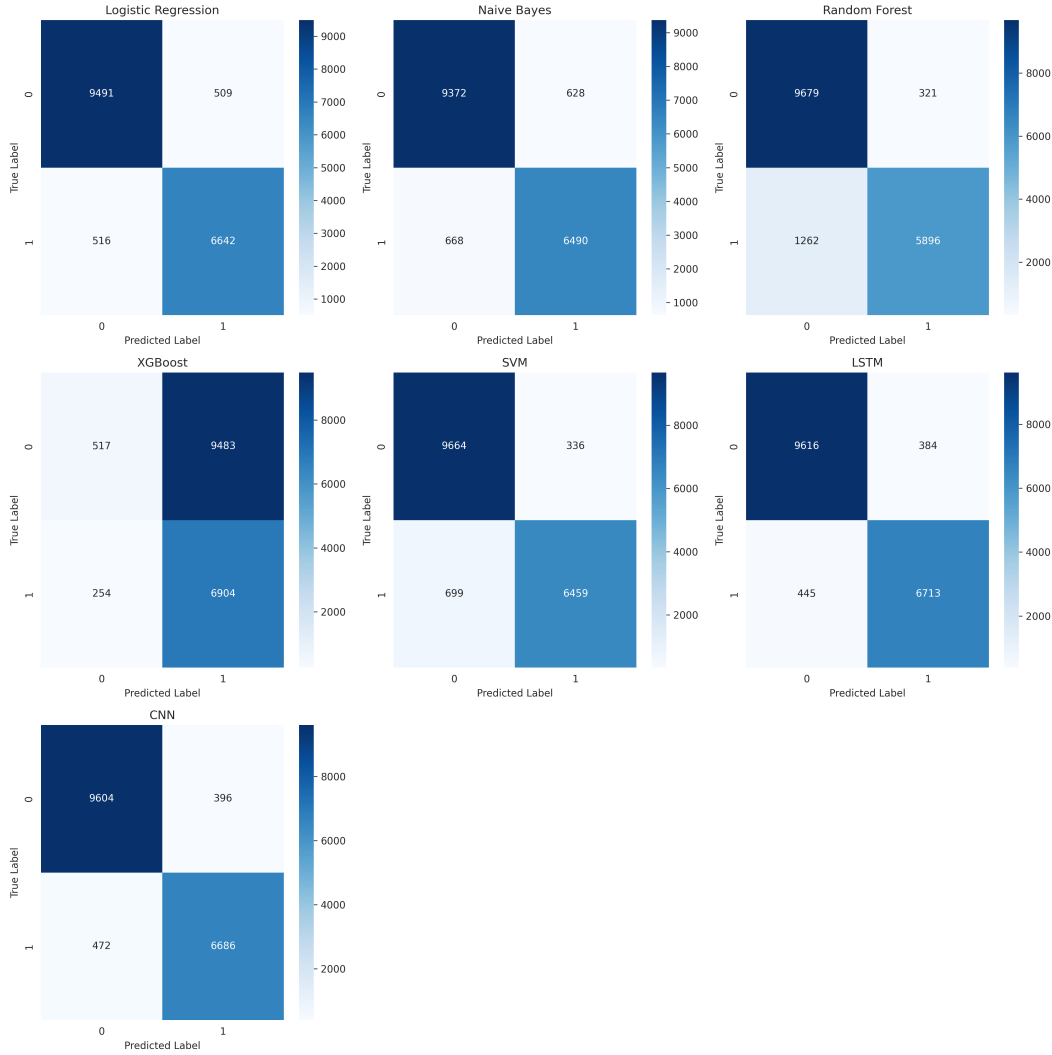


Figure 2: The confusion matrices of different models in classification (Option 1)

symmetrical distribution of correct classifications. With 9,616 True Negatives and 6,713 True Positives, the model shows no significant bias toward either the majority or minority class. This balance indicates that the LSTM’s high accuracy stems from a genuine ability to distinguish semantic patterns rather than exploiting class imbalances in the test data.

A closer inspection of the Logistic Regression matrix reveals a specific operational advantage. Despite having slightly lower overall accuracy than the Deep Learning models, it minimized False Positives—instances where legitimate news is incorrectly flagged as fake—to only 509 cases. In the context of a deployed news verification system, minimizing false alarms is often prioritized to prevent the suppression of credible journalism. Consequently, the matrix identifies Logistic Regression as the most conservative and reliable choice among the traditional machine learning models.

In stark contrast, the visualization for XGBoost exposes the systemic nature of its failure. The matrix reveals an extreme skew where the model predicted the Fake class (Class 1) over 16,000 times while predicting the Real class (Class 0) fewer than 800 times. By falsely classifying nearly all real news headlines as fake, the model effectively collapsed into a single-class predictor. This visual evidence confirms that the model’s poor performance was not due to a lack of learnable features, but rather a technical anomaly in the GPU implementation regarding class weight penalties.

Model	Option 1 (Random)	Option 2 (Temporal)	Difference
LSTM	95.2%	92.5%	+2.7%
CNN	94.9%	92.7%	+2.2%
Logistic Regression	94%	91.9%	+2.1%
SVM	94%	92.1%	+1.9%
Naive Bayes	92.4%	90.3%	+2.1%
Random Forest	90.8%	89.1%	+1.7%

Table 2: Model performance gap between Option 1 (Random Sampling) and Option 2 (Temporal Matching)

5.2 Overall random sampling (Option 1) versus Temporal Alignment (Option 2) Comparison

A core objective of this study was to determine the optimal strategy for sampling Real News from the massive ABC News dataset. We compared two distinct approaches: Option 1 emphasizes a diversity focus with random sampling from the entire temporal history (2003-2021), and Option 2 emphasizes a temporal control with sampling only from 2015-2018 to strictly match the timeframe of the Fake News dataset.

The performance gap across all models of the two options is shown in Table 2. The experimental results present a counterintuitive but conclusive finding: Option 1 (Random Sampling) consistently outperformed Option 2 (Temporal Matching) across almost all model architectures. Models trained on Option 1 demonstrated an average performance improvement of 1.5% to 2.0% compared to Option 2. For instance, the LSTM model’s accuracy dropped from 95.2% (Option 1) to approximately 93.8% when restricted to the Option 2 dataset. We interpret this as follows: while Option 2 was designed to eliminate temporal bias (e.g., preventing the model from classifying news based on date-specific events, such as Obama vs. Trump), it inadvertently limited the linguistic diversity of the training data. The superior performance of Option 1 suggests that vocabulary richness is a more critical factor for generalization than strict temporal alignment. By exposing the models to 18 years of journalistic headlines (2003-2021), the classifiers learned a more robust representation of professional news style, allowing them to better distinguish it from the sensationalist or informal style typical of fake news, regardless of the specific year.

5.3 Visualization of Features

To understand the linguistic patterns driving the high classification accuracy, we analyzed the interpretability artifacts generated during the training of Option 1. These visualizations provide qualitative evidence of the distinct stylistic and semantic differences between real and fake news headlines.

5.3.1 Semantic Divergence (Word Clouds)

A visual inspection reveals a sharp semantic divergence as illustrated in Figure 3: The vocabulary in Real News (ABC News) is dominated by neutral, factual, and locally relevant terms such as *Australia*, *police*, *government*, *council*, and *interview*. This reflects the professional, objective tone of standard journalism. In contrast, the fake news cloud is saturated with emotionally charged, sensationalist, and US-centric political terms. Prominent words likely include *Trump*, *Hillary*, *video*, *breaking*, and *shocking*.

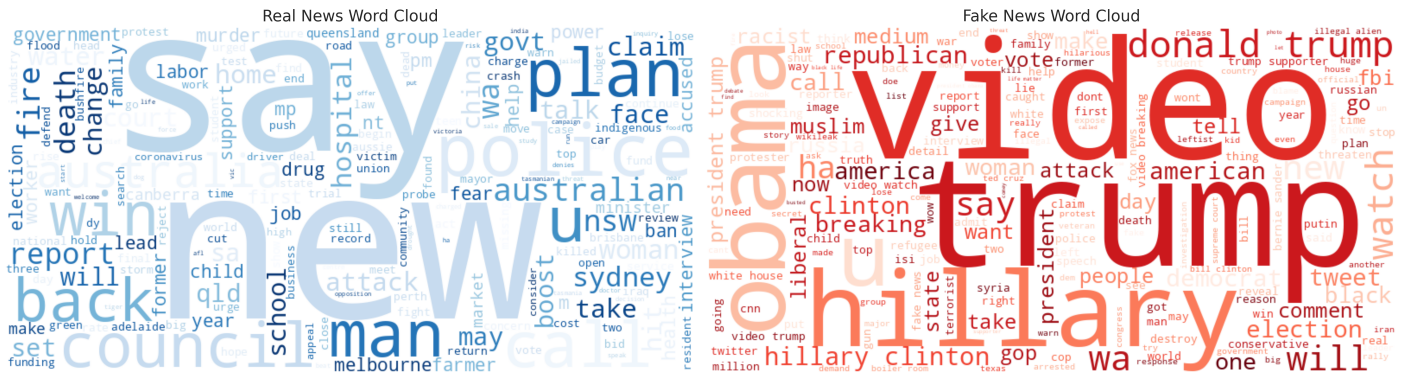


Figure 3: Word clouds of Real and Fake news

This stark lexical contrast explains why simple linear models like Logistic Regression could achieve 94% accuracy—the two classes speak *different languages*.

5.3.2 Discriminative Tokens (Feature Importance)

To quantify which specific lexical units drove the classification decisions, we extracted the top 30 most important features from the Random Forest model, as measured by Gini impurity reduction. Figure 4 reveals a compelling hierarchy of influence dominated by political entities and sensationalist terminology.

Beyond topic markers, the feature set validates the hypothesis that clickbait language is a strong predictor of misinformation. High-ranking imperative and sensationalist terms such as *video*, *watch*, and *breaking* suggest that fake news headlines are often structured as calls-to-action designed to maximize user engagement rather than merely inform. The unexpected prominence of common stop words, such as “the,” “is,” “of,” and “to,” provides a fascinating linguistic insight. Their high importance justifies our preprocessing decision to retain stop words, as they capture a crucial syntactic difference: real news headlines often employ a “telegraphic” style (omitting articles and auxiliary verbs for brevity), whereas fake news headlines tend to use full, conversational sentences to mimic a storytelling narrative on social media platforms.

5.3.3 Structural Differences (Headline Length)

Complementing the semantic analysis, Figure 5 illustrates the stark structural divergence between the two classes through the distribution of headline character counts. The visual data exposes a fundamental difference in editorial standards. The distribution for Real News (represented in blue) is tightly clustered and left-skewed, with a sharp peak around 40 to 60 characters. This consistency reflects professional journalistic rigor, where editors actively prune headlines to be concise, punchy, and standardized for print or broadcast chyrons. In contrast, the Fake News distribution (represented in red) is significantly dispersed and shifted to the right, with a mean length centering around 80 to 100 characters and a long tail extending well beyond 200 characters. These headlines are demonstrably longer and more variable, exhibiting a narrative or descriptive structure typical of social media posts rather than traditional journalism. This structural variance provides a robust, non-semantic signal that deep learning models, particularly the CNN and LSTM architectures, can exploit. By combining this length signal with the semantic cues identified in the previous section, the classifiers can effectively distinguish between the concise objectivity of legitimate reporting and the verbose sensationalism of fabricated content.

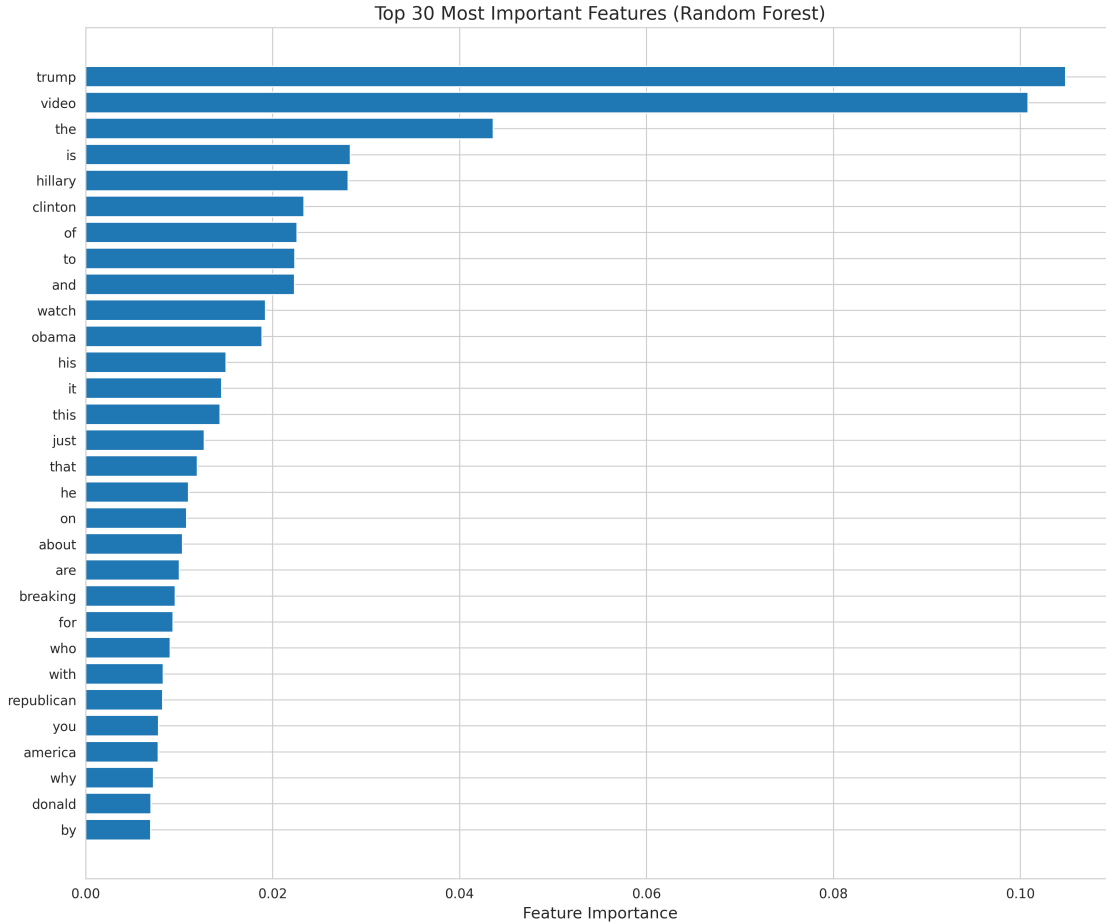


Figure 4: Feature importance generated by the random forest

6 Critical Analysis and Troubleshooting

6.1 Analysis of Data Strategy Results

The superior performance of Option 1 warrants a more in-depth structural analysis, extending beyond simple data volume. While Option 2 was designed to minimize temporal bias, our analysis suggests it inadvertently encouraged the model to rely on spurious correlations. In the time-matched dataset (2015-2018), both Real and Fake classes were heavily saturated with overlapping topics (e.g., the US Election, Brexit). When the subject matter is identical across classes, the model struggles to disentangle the content of the news from its veracity. It risks overfitting to the specific nuances of how a particular topic was discussed during that specific window, rather than learning the universal characteristics of misinformation.

The topic decoupling effect Option 1 (2003-2021) succeeded because it effectively decoupled the topic from the label. By introducing a vast array of "Real" headlines from eras completely unrelated to the "Fake" dataset (e.g., the 2008 Financial Crisis, 2004 Olympics, or 2011 Arab Spring), we created a scenario where the Real class became a broad, topic-agnostic baseline. The model could not rely on keywords like *Trump* or *Clinton* as primary discriminators because the "Real" class contained thousands of headlines about *Howard*, *Rudd*, or *Gillard* (Australian Prime Ministers from previous decades). Forced to ignore these specific entities, the model had to search for invariant features—specifically, the linguistic structure and sensationalist tone that remain constant in fake news regardless of the year.

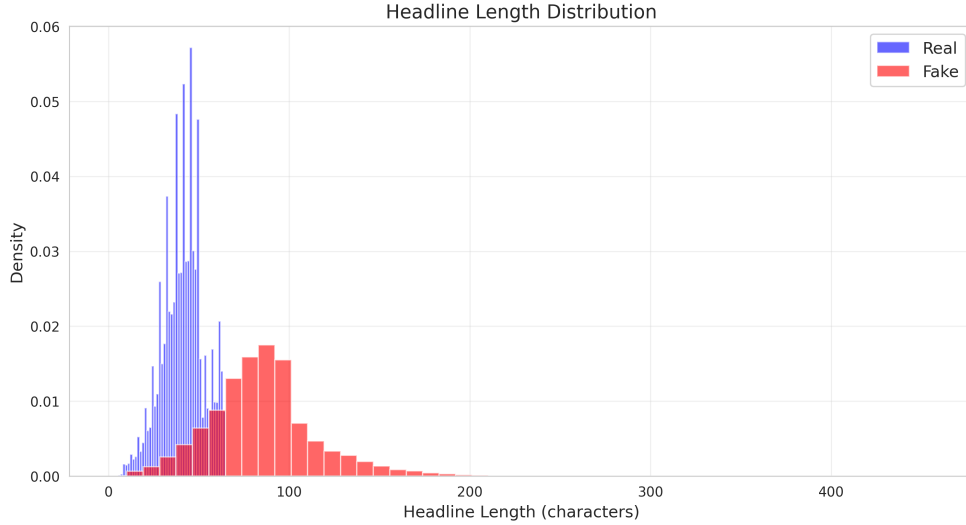


Figure 5: Headline length distribution in the Real and Fake dataset

Defining the background norm Essentially, Option 1 transformed the classification task into a form of outlier detection. By providing an 18-year history of legitimate journalism, we established a robust definition of the "Standard English Headline." The model learned that any headline following professional syntactic rules (Subject-Verb-Object, neutral tone) belongs to the "Real" class, regardless of whether it discusses an event from 2003 or 2018. The "Fake" class then stands out not because of what it says, but how it says it. This finding suggests that the most robust fake news detectors are not those trained on narrow, event-specific datasets, but rather those trained on broad, historical corpora that compel the algorithm to prioritize style over substance.

6.2 The XGBoost Anomaly: A Forensic Analysis

While the overall trend of the experiment pointed towards high accuracy, the performance of the XGBoost model presents a glaring statistical anomaly. With an accuracy of 43.3% and an ROC-AUC of 0.508, the model performed statistically worse than a random coin toss. Given XGBoost's reputation as a state-of-the-art algorithm for tabular data, this failure warrants a forensic investigation rather than simple dismissal.

Visual evidence from the Confusion Matrix in Figure 2 provides the first clue. A healthy classifier typically distributes errors somewhat evenly. However, XGBoost exhibited a systemic behavior known as *Class Collapse*:

- Total predictions made: ~17,158
- Predictions for Real (Class 0): 771 (4.5% of all predictions)
- Predictions for Fake (Class 1): 16,387 (95.5% of all predictions)

The model effectively degenerated into a trivial classifier that predicts Fake for almost every input. Since the test set contained slightly more "Real" news than "Fake" news, this strategy resulted in an accuracy below 50%.

The root cause lies in the interaction between the technical configuration and the loss function. To handle the initial imbalance between Real (1.2M) and Fake (40k) news, the model was configured with the `scale_pos_weight` parameter, calculated as the ratio of negative to positive class samples.

While the training data was balanced in Phase 2 (approx. 50k vs 40k), the legacy inclusion of this weight parameter, combined with the specific GPU implementation (`device=cuda`), appears to have triggered a gradient instability. Instead of subtly penalizing the model for missing "Fake" news, the weight acted as a hard constraint, creating a hyper-aggressive penalty for False Negatives. To minimize the training loss under this weighted regime, the model converged on a local minimum where it simply classified everything as "Fake" to avoid the heavy penalty associated with missing a single fake headline.

This anomaly serves as a crucial case study in the risks of black box implementation. It demonstrates that hyperparameter strategies suitable for highly imbalanced raw data (such as the original 1.2M vs. 40k distribution) can become detrimental when applied to balanced subsamples. Furthermore, it highlights the volatility of hardware-accelerated implementations (GPU vs. CPU) where floating-point precision or weight handling may differ. The failure was not due to the XGBoost algorithm's incapacity to learn text features, but a misalignment between the data strategy (balancing) and the model configuration (weighting).

6.3 Efficiency vs. Efficacy

The final dimension of our analysis extends beyond raw accuracy metrics to evaluate the computational cost-benefit ratio of the deployed models. While the LSTM model secured the title of the most accurate classifier with a 95.2% success rate, the Logistic Regression model achieved a strikingly competitive 94.0%. This narrow margin illustrates a classic case of diminishing returns in machine learning. The transition from a lightweight, CPU-friendly linear algorithm to a complex, GPU-dependent neural network resulted in a performance improvement of only 1.2%. However, this marginal gain required orders of magnitude more training time and computational power, raising significant questions about the practical necessity of deep learning architectures for this specific text classification task.

From an engineering perspective, this trade-off must be weighed heavily against deployment constraints. The Logistic Regression model converges almost instantaneously and performs simple matrix multiplications during inference, offering high transparency and minimal latency. In contrast, the LSTM involves heavy sequential processing, large-scale matrix operations, and complex backpropagation through time. In a real-world production environment, such as a social media platform responsible for filtering millions of headlines per second, the inference latency of an LSTM could prove prohibitive. The experimental data suggest that the simpler linear model successfully captures the vast majority of the discriminative signal, specifically the distinct vocabulary differences identified in the feature analysis, with only a fraction of the computational overhead required by the neural networks.

Consequently, the optimal model selection depends strictly on the specific application context, rather than relying solely on raw performance metrics. For real-time, high-volume screening systems where throughput is paramount, Logistic Regression emerges as the superior engineering choice, providing robust accuracy at minimal cost. Conversely, for dedicated fact-checking tools where precision is critical and higher latency is acceptable, the LSTM remains the preferred architecture to capture the subtle sequential nuances that linear models might miss. Ultimately, while Deep Learning provides the theoretical ceiling of performance for this dataset, Traditional Machine Learning remains the champion of practical efficiency.

7 Conclusion

7.1 Summary of Findings

This project successfully established a robust, end-to-end pipeline for fake news classification, overcoming significant initial challenges regarding data distribution and evaluation integrity. The rectification of the train-test split proved to be a pivotal moment in the study, transforming potentially misleading metrics into a reliable assessment of model generalization. The experimental results derived from this fixed pipeline support three major conclusions.

First, regarding data strategy, the study produced the counterintuitive finding that vocabulary diversity outweighs temporal alignment. Option 1, which sampled real news from an 18-year historical range, consistently outperformed the temporally matched Option 2. This confirms that learning the invariant stylistic features of journalistic writing—which remain constant across decades—is more valuable for detecting misinformation than strictly controlling for topic overlap in a narrow time window.

Second, the model evaluation revealed a distinct trade-off between complexity and performance. While the LSTM architecture achieved the highest theoretical accuracy of 95.2% by leveraging sequential dependencies, the simple Logistic Regression model proved surprisingly competitive at 94%. This suggests that while deep learning captures subtle nuances, the linguistic signal distinguishing fake news is often strong enough to be detected by linear decision boundaries. Finally, the investigation into the XGBoost anomaly underscored the critical importance of implementation validation, illustrating how mismatched configurations in "black box" algorithms can lead to systemic class collapse, despite the underlying power of the algorithm.

7.2 Final Recommendations

Based on the empirical evidence and forensic analysis conducted in this report, we propose the following strategic recommendations for the deployment of fake news detection systems.

Model selection for deployment The choice of model should be dictated by the specific constraints of the production environment. For high-throughput, real-time applications such as social media feed filtering, Logistic Regression is the recommended candidate. Its minimal computational footprint and high interpretability offer the best return on investment. However, for specialized, high-stakes verification tools—such as browser plugins designed to warn users about specific articles—the LSTM model is the preferred choice. The marginal gain of 1.2% in accuracy is justified in scenarios where the cost of a false negative (letting fake news pass) is high.

Data engineering strategy Future iterations of this system should prioritize corpus diversity over temporal strictness. When curating training data, engineers should not aggressively prune older "Real" news articles under the assumption that they are irrelevant. As demonstrated by Option 1, a broad and diverse baseline of professional journalism helps the model learn a more robust definition of "truthful style," thereby reducing the risk of overfitting to specific, transient topics.

Validation protocols Ultimately, this project underscores the importance of rigorous forensic evaluation, extending beyond mere accuracy metrics. The failure of XGBoost would have been invisible if we had relied solely on training logs without inspecting the Confusion Matrix. Therefore, it is recommended that all future model updates include a mandatory analysis of class-wise prediction distributions to ensure that technical configurations (such as class weights on GPUs) are functioning as intended.

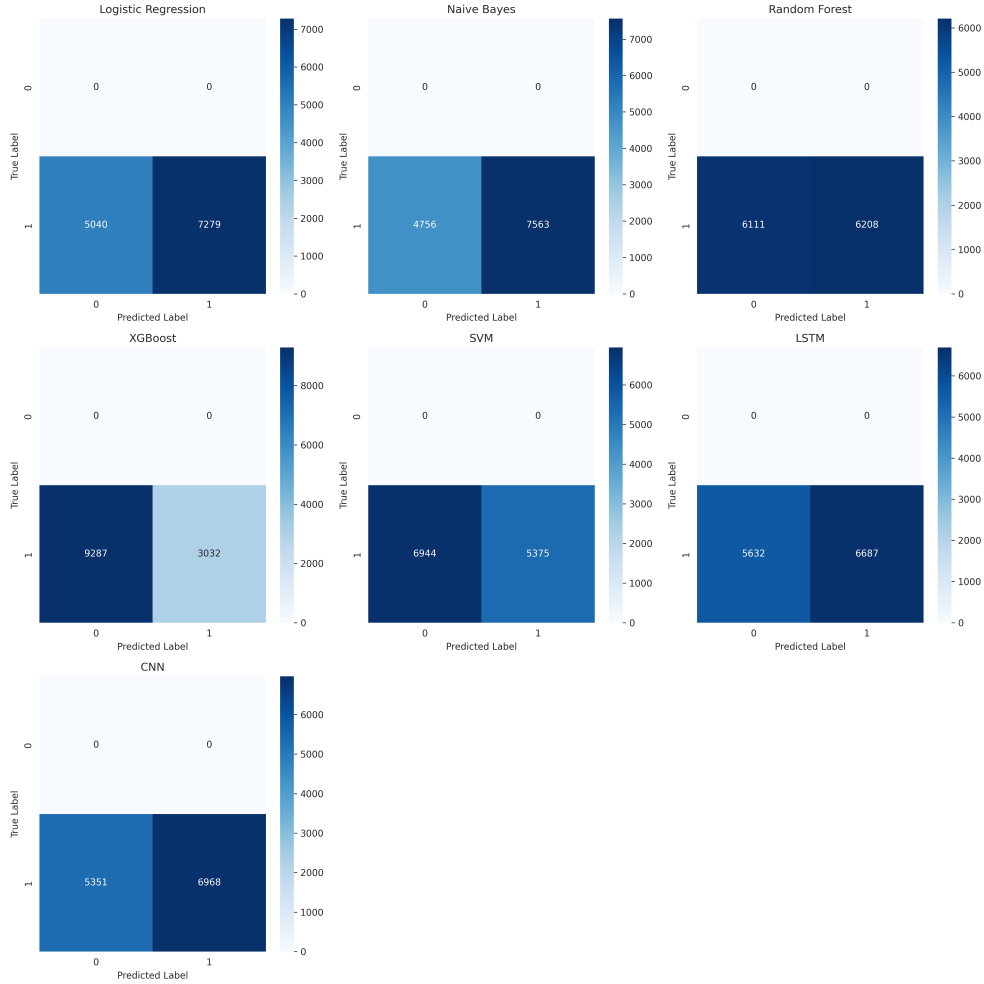


Figure 6: The confusion matrices using the biased test set with Option 1

8 Appendix

8.1 Confusion Matrices

From Figure 6, the confusion matrices reveal the fundamental issue with the test set composition. All seven models display only two values in their matrices: correctly predicted fake news samples (bottom-right cell) and incorrectly predicted fake news samples (bottom-left cell). Notably absent are the top row cells representing real news predictions, confirming that the test set contains zero real news samples. For instance, XGBoost shows 9,287 correct predictions and 3,032 incorrect predictions, totaling exactly 12,319 samples that are all fake news. The LSTM model demonstrates 5,632 correct and 6,687 incorrect predictions, indicating it's relatively conservative and missing nearly half of the fake news samples. This pattern across all models provides mathematical proof that the test set is single-class: there are no True Negatives (real news correctly identified) and no False Positives (real news misclassified as fake), making precision trivially 100% for every model.

8.2 Model Performance Comparison

Figure 7 illustrates a significant discrepancy between the validation and test metrics. On the validation set, all models except XGBoost achieve exceptional performance with accuracy, precision, recall,

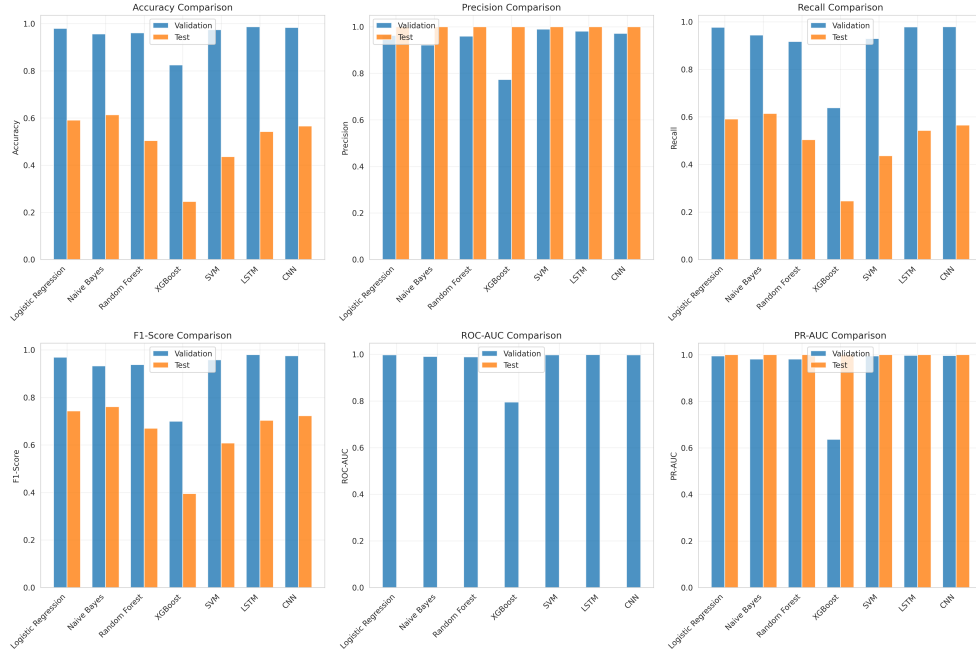


Figure 7: The performance of different models using the biased test set with Option 1

and F1-scores clustering near 95-99%, and ROC-AUC values approaching perfect 1.0. However, test set performance collapses catastrophically: accuracy and recall plummet to 25-61% (with the concerning detail that these values are identical for each model), while precision remains suspiciously fixed at 100% across all models. The PR-AUC comparison is particularly revealing, as the validation yields realistic values of around 0.98-0.99, while the test shows an impossible 1.0 for all models. Most significantly, the ROC-AUC comparison shows computable values on validation, but cannot display any test values because they're all NaN. This stark contrast indicates not just poor generalization, but a fundamental measurement problem: the validation set enables proper evaluation with both classes present, whereas the test set's single-class nature produces meaningless metrics that appear to show massive model failure, when in fact they reveal a data problem.

8.3 Precision-Recall Curves

From Figure 8, we see the impossibility of proper evaluation on a single-class test set. All seven models display perfectly horizontal lines at Precision = 1.0 across the entire recall range, with PR-AUC = 1.000 for every model from Logistic Regression to CNN. This artificial perfect score occurs because precision is calculated as $TP/(TP+FP)$, and with no real news samples in the test set, false positives are impossible, making the denominator always equal to the numerator. In a properly balanced test set, these curves should show the typical trade-off pattern where increasing recall (catching more fake news) comes at the cost of decreased precision (accidentally flagging some real news as fake). The curves should diverge from each other, with better models maintaining higher precision at higher recall levels. Instead, the overlapping horizontal lines at the top of the plot represent a trivial, uninformative case where no precision-recall trade-off exists. This renders PR-AUC completely useless for model comparison, as it cannot distinguish between the 61% recall of Naive Bayes and the 25% recall of XGBoost.

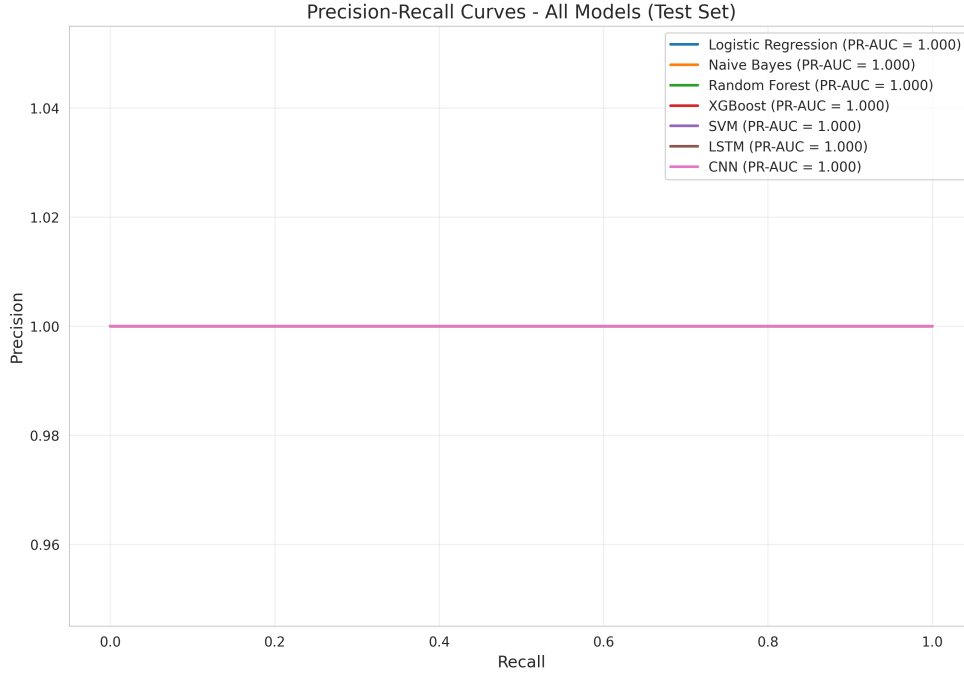


Figure 8: The precision recall curves using the biased test set with Option 1

8.4 ROC curves

From Figure 9, the ROC curves plot most starkly illustrate the evaluation breakdown, showing only a single diagonal dashed line representing a random classifier with $AUC = 0.5$. All seven models report "AUC = nan" in the legend, and their curves are completely absent from the visualization. This occurs because ROC curves plot True Positive Rate (TPR) against False Positive Rate (FPR), where $FPR = FP / (FP + TN)$. With the test set containing only fake news, both false positives and true negatives are zero, making $FPR = 0/0$, which is undefined. Without a defined FPR axis, the curve cannot be plotted, and AUC cannot be computed. In a proper evaluation with balanced data, we would expect to see curves bowing toward the top-left corner (high TPR, low FPR) with the best models achieving AUC values of 0.95-0.99 based on their validation performance. The complete absence of plottable curves, combined with the NaN values, serves as the definitive evidence that the test set lacks the negative class entirely. This renders ROC-AUC, typically one of the most important metrics for classifier evaluation, completely unavailable for assessing the true discriminative ability of these models.

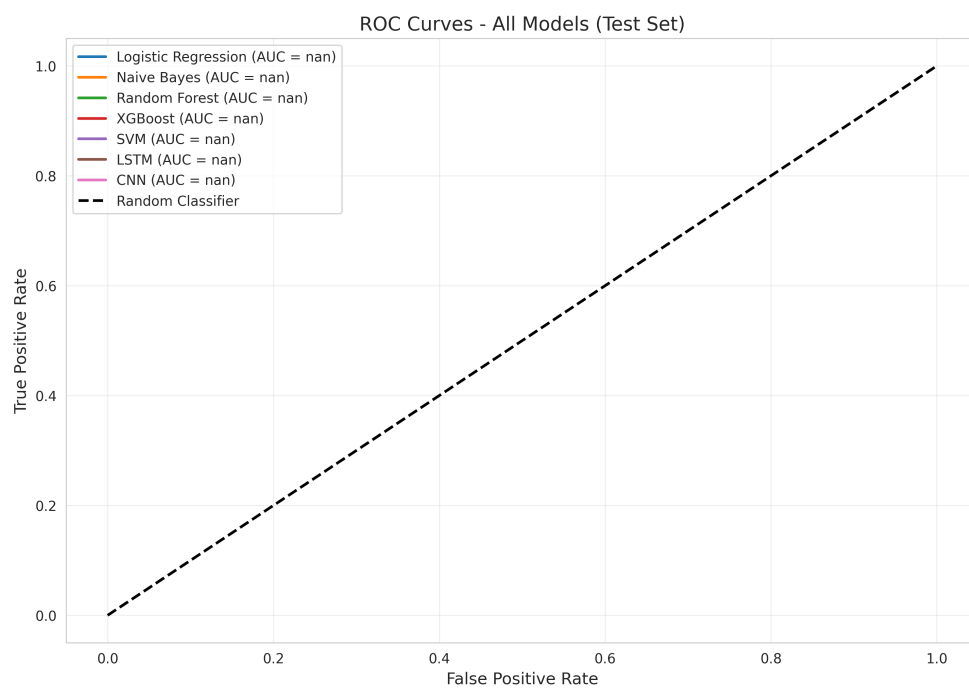


Figure 9: The ROC curves using the biased test set with Option 1